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United States Patent Application Publication

20250268115

Kind Code

A1

Publication Date

August 28, 2025

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Gardening or gardening or husbandry tool and related method

Abstract

Disclosed is an integrated rake and hoe tool.

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Family ID: 1000007852828

Appl. No.: 18/588619

Filed: February 27, 2024

Publication Classification

Int. Cl.: A01B1/20 (20060101); A01B1/10 (20060101); A01D7/02 (20060101)

U.S. Cl.:

CPC A01B1/20 (20130101); A01B1/10 (20130101); A01D7/02 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not applicable.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable.

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not applicable.

REFERENCE TO AN APPENDIX SUBMITTED ON A COMPACT DISC AND

INCORPORATED BY REFERENCE OF THE MATERIAL ON THE COMPACT DISC

[0004] Not applicable.

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR A JOINT INVENTOR

[0005] Reserved for a later date, if necessary.

BACKGROUND OF THE INVENTION

Field of Invention

[0006] The disclosed subject matter is in the field of gardening or gardening or husbandry tools.

Background of the Invention

[0007] Thousands of years ago, humans began domesticating animals and cultivating land, leading to the development of various tools to aid in these tasks. Over time, these tools evolved. That said, the fundamental need for efficient, multi-purpose implements has remained constant.

[0008] For example, gardeners and husbandry practitioners frequently find themselves switching between a rake and a hoe to accomplish routine tasks. This constant interchange not only interrupts the workflow but also contributes to physical strain and time inefficiency. A need for a dual-functionality tool emerges as a response to the practical challenges faced by such gardeners and husbandry practitioners.

SUMMARY OF THE INVENTION

[0009] The gardening or husbandry tool **1000** combines the functionalities of a rake and a hoe into a single, versatile tool. This innovative design addresses the need for a more efficient and ergonomic solution for various tasks in gardening, farming, and land management. The gardening or husbandry tool **1000** is defined by two main components: the rake-part **1100** and the hoe-part **1200**. The rake-part **1100** features outward-fanning rake tines **1110**, designed for gathering, leveling, and aerating soil or mulch. The hoe-part **1200**, equipped with cutting blades **1210**, serves as a robust tool for cutting, digging, and lightwork-hoeing, particularly useful for breaking up hardened soil or cutting through small roots and non-wooded weeds.

[0010] A unique aspect of the tool **1000** is the integration of these two parts, allowing the tool to be used as a traditional rake when operated in a forward-to-backward motion and as a hoe or machete when operated side-to-side. This dual functionality not only enhances the tool's versatility but also reduces the need for multiple separate tools, making gardening and farming tasks more efficient and less physically demanding. The integration of the rake and hoe functionalities into a single tool represents a significant innovation in gardening and agricultural tools. By enabling users to seamlessly transition between raking and hoeing without the need to switch tools, the husbandry tool **1000** significantly enhances operational efficiency. This not only saves time but also reduces the physical burden on the user, thereby improving the overall gardening and land management experience.

Description

BRIEF DESCRIPTION OF THE VIEWS OF THE DRAWINGS

[0011] Other objectives of the disclosure will become apparent to those skilled in the art once the invention has been shown and described. The way these objectives and other desirable characteristics can be obtained is explained in the following description and attached figures in which:

[0012] FIG. **1** is a perspective view of an improved gardening or husbandry tool **1000** defined by a rake-part **1100** with rake tines **1110** and a hoe-part **1200** with cutting blades **1210** such that when the two parts **1100**, **1200** are coupled, the gardening or husbandry tool **1000** is defined by a rake with tines **1110**, **1210** that fan outward so that the tool **1000** can be used as a rake when the tool **1000** is operated forward-to-backward and thicker side tines **1210** that can be used as a hoe or machete blades when the tool is operated side-to-side;

[0013] FIG. 2 is a perspective view of the hoe-part **1200** of the gardening or husbandry tool **1000**;
[0014] FIG. 3 is a perspective view of the rake part of the gardening or husbandry tool **1000**;
[0015] FIG. 4 is an exploded view of the gardening or husbandry tool **1000** that depicts coupling of the hoe-part **1200** and the rake part **1100** of the tool **1000**;
[0016] FIG. 5 is an environmental tool that shows the tool **1000** being operated in a forward to backward or side-to-side manner;
[0017] FIG. 6 is a perspective view of a preferred design of the tool **1000**;
[0018] FIG. 7 is a perspective view of another embodiment of an improved gardening or husbandry tool **2000**.

[0019] FIG. 8 is a perspective view of the other embodiment of the tool **2000** with a handle installed;
[0020] FIG. 9 is a perspective view of the tool **2000** with a handle installed along with side-by-side views of the bottom and top of perspective views of the alternative embodiment **2000** of the tool; and,

[0021] FIG. 10 is a zoomed in view of the underside of the tool **2000** and the underside thereof. It is to be noted, however, that the appended figures illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments that will be appreciated by those skilled in the relevant arts. Also, figures are not necessarily made to scale but are representative.

DETAILED DESCRIPTION OF PROFFERED EMBODIMENTS

[0022] Disclosed is an integration of rake and hoe or machete functionalities into a single tool. Generally, the disclosed gardening or husbandry tool is defined by a rake with tines that fan outward so that the tool can be used as a rake when the tool **1000** is operated forward-to-backward and thicker side tines **1210** that can be used as a hoe or machete blades when the tool is operated side-to-side. The more specific details of the disclosure are described in connection with the figures.

[0023] FIG. 1 is a perspective view of an improved gardening or husbandry tool **1000** defined by a rake-part **1100** with rake tines **1110** and a hoe-part **1200** with cutting blades **1210**. FIG. 2 is a perspective view of the hoe-part **1200** of the gardening or husbandry tool **1000**. FIG. 3 is a perspective view of the rake part of the gardening or husbandry tool **1000**.

[0024] Referring to FIG. 2, the hoe-part **1200** is defined by an A-frame. In a preferred embodiment, the A-Frame features a socket **1220** at the top and is constructed such that the legs of the A-frame define blades **1210** of a hoe or machete or else two outer tines of a rake. As discussed later in connection with FIG. 4, the cross-bar of the A-frame provides a rigid backing or back bone for the tines **1110** of the rake part **1100** (See FIGS. 1 or 4).

[0025] Referring to FIG. 3, the rake-part **1100** is defined by an upside down T-frame. In a preferred embodiment, the upside down T-frame features a stem **1120** at the top and is constructed such that the legs of the crossbar of the T-frame define catches **1230** for the coupling to the A-Frame of the hoe-part **1200** (See FIG. 4). Suitably, the cross bar of the T-frame features protruding tines **1110** of a rake.

[0026] The development of the gardening or husbandry tool **1000** involved careful consideration of ergonomics, durability, and functionality. The exploded view (FIG. 4) demonstrates the thoughtful design of the coupling mechanism between the rake-part **1100** and the hoe-part **1200**, ensuring a secure and seamless transition between functions. As shown, the crossbar of the A-frame provides a rigid backing or back bone for the tines **1110** of the rake part **1100** and the catches **1230** of the T-frame secure the two frames in a braided or weaved configuration. Specifically, the stem of the t-frame is provided under the crossbar of the A frame and plugged into the socket of the A-frame while the catches of the T frame are provided over the legs of the A frame and while the tines are positioned under the cross-bar of the A-frame.

[0027] The gardening or husbandry tool **1000** stands as a testament to the evolution of agricultural

tools, reflecting both a deep respect for the traditions of animal husbandry and a forward-looking approach to tool design. FIG. 5 shows that, by combining the functionalities of a rake and a hoe, the tool **1000** offers a versatile solution that addresses the multifaceted demands of contemporary gardening, farming, and land management. As shown, the improved gardening or husbandry tool **1000** is defined by a rake-part **1100** with rake tines **1110** and a hoe-part **1200** with cutting blades **1210** such that when the two parts **1100**, **1200** are coupled, the gardening or husbandry tool **1000** is defined by a rake with tines **1110**, **1210** that fan outward so that the tool **1000** can be used as a rake when the tool **1000** is operated forward-to-backward and thicker side tines **1210** that can be used as a hoe or machete blades when the tool is operated side-to-side.

[0028] A purpose of the improved husbandry tool, designated as tool **1000**, is to provide a versatile and efficient implement for various agricultural and gardening tasks. This tool combines the functionalities of a rake and a hoe into a single device, enhancing the convenience and effectiveness of land management practices. The rake-part **1100**, with its outward-fanning rake tines **1110**, is designed for tasks such as gathering, leveling, and aerating soil or mulch. This part of the tool is used when the tool is operated in a forward-to-backward motion, functioning as a traditional rake. The hoe-part **1200**, featuring thicker cutting blades **1210**, is intended for more robust activities such as cutting, digging, and hoeing. This part is particularly useful for breaking up hard soil or cutting through small roots and non-wooded weeds. When the tool is operated side-to-side, these thicker side tines can be used as a hoe or machete blades. By integrating these two parts, the husbandry tool **1000** allows users to switch between raking and hoeing functions without the need to change tools, thereby saving time and reducing the physical strain associated with carrying and switching between multiple implements. This multifunctional design aims to improve the efficiency of agricultural operations and support the day-to-day care, management, and cultivation of land.

[0029] The preferred design of the tool **1000** (FIG. 6) showcases an optimized form that balances weight, strength, and ease of use. This design is the culmination of extensive research and testing, aimed at providing users with a high-quality, durable tool that meets the diverse needs of modern husbandry and gardening practices. FIG. 7 is a perspective view of an improved gardening or husbandry tool **2000** defined by rake tines **2110** and cutting blades **2210**. FIG. 8 is a perspective view of the other embodiment of the tool **2000** with a handle installed. FIG. 9 is a perspective view of the tool **2000** with a handle installed along with side-by-side views of the bottom and top of perspective views of the alternative embodiment **2000** of the tool. FIG. 10 is a zoomed in view of the underside of the tool **2000** and the underside thereof. As shown in these figures the alternative embodiment of the tool **2000** operates in the same manner as the earlier described embodiment of the tool **1000**.

[0030] As compared to the earlier embodiment **1000** depicted in FIG. 6, the alternative embodiment **2000** of FIGS. 7-10 features several different structural configurations. One difference could be that the alternative embodiment **2000** has a single piece neck support that is defined by folded sheet metal while the earlier embodiment **1000** of the tool of FIG. 6 features a three-piece neck support defined by rivetted sheet metal. Moreover, what was considered a rake part **1100** and a hoe part **1200** of the earlier embodiment are formed as a single unit with dimple welds and/or rivets connecting the Other embodiments include screws connecting the portions two parts together. together as dimple welds can be or become weak.

[0031] Although the method and apparatus is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features, aspects, and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described, but instead might be applied, alone or in various combinations, to one or more of the other embodiments of the disclosed method and apparatus, whether or not such embodiments are described and whether or not such features are presented as being a part of a described embodiment. Thus, the breadth and

scope of the claimed invention should not be limited by any of the above-described embodiments. [0032] Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, should be construed as open-ended as opposed to limiting. As examples of the foregoing: the term “including” should be read as meaning “including, without limitation” or the like; the term “example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof; the terms “a” or “an” should be read as meaning “at least one,” “one or more,” or the like; and adjectives such as “conventional,” “traditional,” “normal,” “standard,” “known,” and terms of similar meaning should not be construed as limiting the item described to a given time period or to an item available as of a given time, but instead should be read to encompass conventional, traditional, normal, or standard technologies that might be available or known now or at any time in the future. Likewise, where this document refers to technologies that would be apparent or known to one of ordinary skill in the art, such technologies encompass those apparent or known to the skilled artisan now or at any time in the future.

[0033] The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to,” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases might be absent. The use of the term “assembly” does not imply that the components or functionality described or claimed as part of the module are all configured in a common package. Indeed, any or all the various components of a module, whether control logic or other components, might be combined in a single package or separately maintained and might further be distributed across multiple locations.

[0034] Additionally, the various embodiments set forth herein are described in terms of exemplary block diagrams, flow charts, and other illustrations. As will become apparent to one of ordinary skill in the art after reading this document, the illustrated embodiments and their various alternatives might be implemented without confinement to the illustrated examples. For example, block diagrams and their accompanying description should not be construed as mandating a particular architecture or configuration.

[0035] All original claims submitted with this specification are incorporated by reference in their entirety as if fully set forth herein.

Claims

1. A gardening or husbandry tool **1000** defined by two main components: a. the rake-part **1100** and b. the hoe-part **1200**, wherein the rake-part **1100** features outward-fanning rake tines **1110**, designed for gathering, leveling, and aerating soil or mulch, wherein the hoe-part **1200** is equipped with cutting blades **1210** for cutting, digging, and hoeing.
